

HARISH RAGAVENDRA CHITTIBABU

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EDUCATION

Dublin City University

Dublin, Ireland

MSc in Computing(Artificial Intelligence)

Sep 2025 – Aug 2026

- **Thesis:** Explainable AI Tomato Disease Diagnosis with Agentic Reasoning ([link](#)).
- **Related Coursework:** Machine Learning, Artificial Intelligence, Machine Translation, Data Analytics & Data Mining, Large Language Model, RAG, Statistical Data Analysis, Data Management & Visualisation.

B S Abdur Rahman Crescent Institute of Science and Technology

Chennai, India

B.Tech Computer Science Engineering

Oct 2021 – May 2025

- **Thesis:** AI-Driven Smart Robotic System for Sustainable Greenhouse Management
- **Related Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Embedded Systems, Cloud Computing, Machine Learning.

SKILLS

Programming Language: python, JavaScript, SQL, C++.

APIs & Frameworks: RESTful API design, FastAPI, Microservices Architecture.

Databases: PostgreSQL, MySQL, MongoDB, Oracle, Snowflake, Pinecone (vector DB)

Database Engineering: Schema design, query optimisation, CTEs, window functions, recursive queries, orchestration.

Containers & Infrastructure: Docker, Docker Compose, Kubernetes (AWS EKS), Terraform

Cloud: AWS (S3, Lambda, EMR, SageMaker, Bedrock), Azure (Container app, Databricks, Fabric)

Data & Messaging: Kafka, PySpark, ETL pipelines

DevOps & Testing: CI/CD (GitHub Actions), automated testing (PyTest)

Security & Compliance: GDPR-compliant data design, HMAC-SHA256 pseudonymisation, data retention policies.

AI Assist Tools: Kiro, Antigravity, Copilot

EXPERIENCE

Tanama Software Limited

Bengaluru, India

Software Engineering(Remote, Concurrent with studies)

Mar 2023 – Apr 2026

- Debugged and tested the backend of an internal data-storage application holding financial data from Indian banks, building debug APIs, monitoring load and managing caching to keep the service stable under production traffic.
- Built and owned an end-to-end data pipeline processing 200+ GB/day, scalable to 500 GB/day, with health checks, 400 GB volume alerts and PostgreSQL backups on Kubernetes using WAL archiving and point-in-time recovery.
- Designed and built the pipeline that migrated banking data from on-premise servers to AWS S3, with processing on EC2 and a failover cluster that kept APIs and data transfer running with under 30 minutes of downtime.
- Automated validation of the data flow, queries and pipeline outputs, cutting the full verification cycle from 1 week to under 2 days.
- Built a fraud-detection model using gradient boosting (XGBoost, LightGBM) on AWS, scoring thousands of transactions per minute and cutting false positives by 87% versus the previous Spark rule-based system.

Diplomatic Consultancy Services

Muscat, Oman

AI/ML Internship (Remote, Concurrent with studies)

Jun 2022 – Nov 2022

- Built ESP32 and Raspberry Pi camera rigs to capture field images of eggplant and okra plants, creating the dataset for automated crop-damage detection.
- Trained a ResNet model to classify healthy, damaged and pest-infested eggplant and okra, reaching 91% accuracy in pilot deployments.
- Connected the edge devices to cloud service like AWS S3 so captured images flowed automatically into the model for real-time monitoring.

PROJECTS

Explainable AI Tomato Disease Diagnosis with Agentic Reasoning (MSc Thesis, DCU)

Python · PyTorch · ResNet18 · Grad-CAM · LIME · SHAP · LLaMA3 (Ollama) · MarianMT · Gradio

- Built a bilingual (EN/GA) tomato disease diagnosis system using a ResNet18 transfer-learning model (frozen backbone, 5,130 trainable parameters), reaching 92.3% accuracy (Macro-F1 0.903) on 18,160 images across 10 classes with just 20 minutes of CPU-only training.
- Benchmarked Grad-CAM, SHAP and LIME on localisation quality, selecting Grad-CAM (91.3% Center Focus Score) to generate farmer-facing reports with zoomed lesion panels, differential diagnosis and urgency-coded advice.
- Engineered a 5-stage agentic reasoning pipeline in **LLaMA3 (Assess → Locate → Contextualise → Advise → Escalate)** that grounds advice in a verified knowledge base, escalating to an expert automatically and falling back to rule-based templates when offline.
- Integrated English→Irish MarianMT translation, validated on 1,800+ curated agricultural terms, into a web app, achieving **BLEU 52.3 / chrF 61.8** and running the full pipeline in under 5 seconds on CPU.

AML Transaction Monitoring & Alert Triage System

Python · PostgreSQL · FastAPI · Scikit-learn · NetworkX · Plotly Dash · Docker · GitHub Actions

- Built a production-grade AML detection pipeline on 19M synthetic transactions, combining SQL typology rules (CTEs, window functions, recursive queries), Isolation Forest anomaly scoring and NetworkX graph analysis, cutting a raw 84% flag rate to a 0.9% actionable alert rate.
- Designed a GDPR-compliant PostgreSQL schema with HMAC-SHA256 pseudonymisation, EXPLAIN ANALYZE-tuned indexed queries and a data retention policy aligned with UK GDPR, MLR 2017 and POCA 2002, covering the full compliance lifecycle from ingestion to audit.
- **Delivered** a FastAPI REST API (6 endpoints) and a 4-page Plotly Dash dashboard surfacing risk scores and LLM-generated SAR narratives to analysts, **deployed to Azure Container Apps** with Docker Compose and a GitHub Actions CI/CD pipeline.